

Journalizing Automation

Based on OCR and sLLM

Team 3

Principles of Accounting Project

1. Problem

Current Problems in Accounting

- Manual journalizing takes too much time
- Human errors can occur during repetitive work
- Companies process many invoices and receipts every day
- Repetitive accounting tasks reduce efficiency

2. Why OCR + sLLM?

Two Technologies, Two Roles

- **OCR** reads text from receipt or invoice images
- **sLLM** understands the transaction context
- Together, they can generate draft journal entries

Goal: support accountants, not replace them

3. System Structure

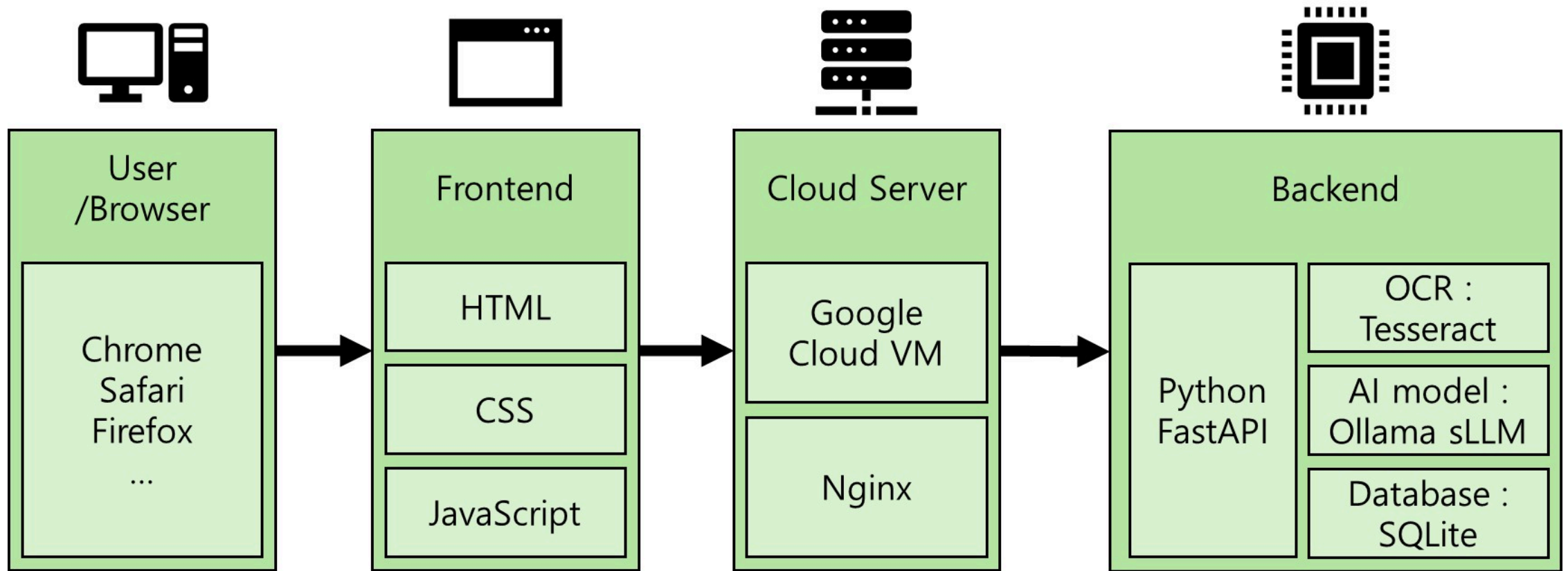
Main Parts of Our Project

- **Frontend:** user upload screen and result display
- **Backend:** OCR, AI inference, validation, and saving
- **Deployment:** cloud server for browser access

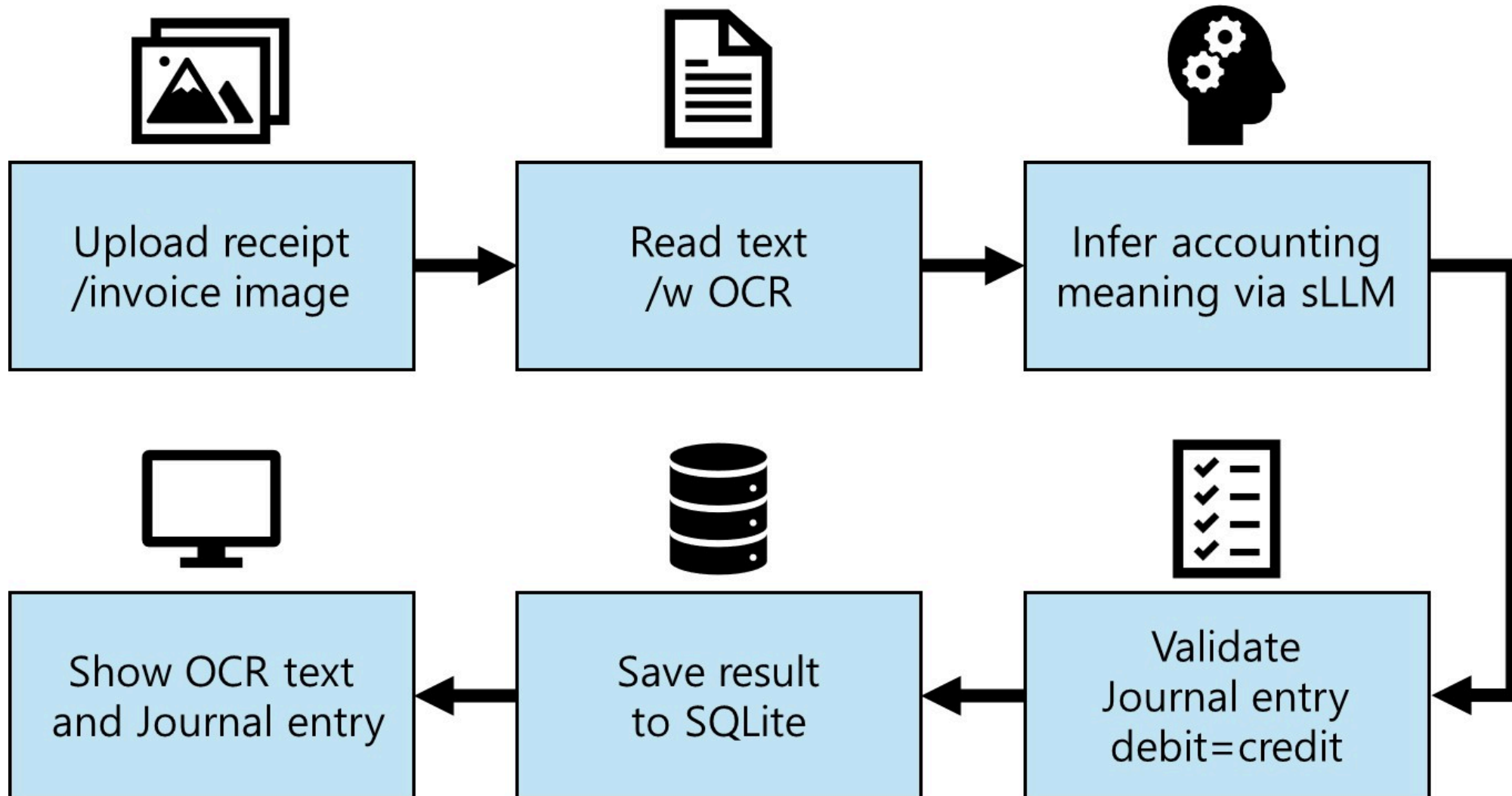
Technology Stack

HTML / CSS / JavaScript / FastAPI / Tesseract / Ollama / SQLite / Google Cloud VM

3. System Structure



4. Data Flow



5. Demo Result

Repair Invoice Example

- Uploaded an English repair invoice
- OCR extracted invoice text clearly
- sLLM generated a journal entry draft
- Backend validated debit and credit equality

Account	Debit	Credit
Office Supplies Expense	154.06	0
Accounts Payable	0	154.06

6. Accounting Concepts

Concepts Used in Our Project

- **Accounting transaction**
 - The event affects accounting elements and can be measured in money
- **Realization of transaction**
 - The system checks whether the transaction is supported by evidence
- **Journaling**
 - The final output is a debit and credit journal entry draft

7. Future Usage

Practical Usefulness

- Reduce time spent on repetitive journalizing
- Help accountants review drafts instead of entering everything manually
- Allow junior accountants to focus on more meaningful tasks
- Support small businesses with limited accounting resources

8. Limitations

Current Limits of Our Project

- AI results still need human review
- OCR quality depends heavily on image quality
- Blurry or small text can lead to incorrect results
- AI mistakes can be risky for important transactions

Current role: assistant tool, not a fully autonomous accounting system

9. Future Work

Possible Improvements

- Improve OCR accuracy
- Strengthen AI validation
- Support multilingual invoice processing
- Add more accounting features
- Expand toward a small AI-based ERP system

Q&A
